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INTERVENTION, EVALUATION, AND POLICY STUDIES

The Effects of the Louisiana Scholarship Program on Student Achievement and College Entrance

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ABSTRACT

The Louisiana Scholarship Program (LSP) offers publicly funded vouchers to moderate- and low-income students in low-performing public schools to enroll in participating private schools. Established in 2008 as a pilot program in New Orleans, the LSP expanded statewide in 2012. Drawing upon the random lotteries that placed students in LSP schools, we estimate the causal impact of using an LSP voucher to enroll in a private school on student achievement on the state accountability assessments in math, English Language Arts, and science over a four-year period, as well as on the likelihood of enrolling in college. The results from our primary analytic sample indicate substantial negative achievement impacts, especially in math, that diminish after the first year but persist after four years. In contrast, when considering the likelihood of students entering college, we observe no statistically significant difference between scholarship users and their control counterparts.

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Introduction

Private school choice continues to be a heavily debated education reform. School choice programs give parents the opportunity to select a school for their children other than their residentially assigned public school. Private school choice, in the form of vouchers, tax-credit scholarships, or Education Savings Accounts (ESAs), provides families financial support to pay tuition for a private school of their choice. Most of the research and policy focus to date has been on the school voucher form of private school choice, funded and administered by state governments. While economist Milton Friedman (1955) introduced the educational voucher concept in the United States, the theoretical support for its desirability dates to political philosophers Thomas Paine (1791) and John Stuart Mill (1962). School voucher theory holds that government should provide funds supporting compulsory education, but it need not necessarily deliver the schooling itself (Friedman, 1955). In theory, vouchers benefit students by facilitating the matching of

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student needs to specific school programs and environments, and by increasing the competitive pressures schools face in the broader education system (DeAngelis & Erickson, 2018; Moe, 1995; Viteritti et al., 2005).

The theoretical claim that school vouchers will improve outcomes for students has been widely contested. Voucher skeptics argue that student background largely determines outcomes, so providing opportunities for students to attend a different school without changing other substantial living conditions will have little positive effect on student achievement (Condliffe et al., 2015). Other critics worry that parents will choose poorly when using a voucher to select a school for their child (Lauder & Hughes, 1999; Smith & Meier, 1995). A third claim is that private school voucher programs will generate negative outcomes by injecting competition into an environment where cooperation is essential for student success (Abrams, 2016; Bryk et al., 1993). A fourth claim is that private schools will not teach the same subjects as public schools in the same grades, especially in the sensitive area of civics, making it difficult for students to switch school sectors without learning loss and leaving privately-schooled adults unprepared for their responsibilities as citizens (Gutmann, 1987; Venegoni & Ferrero, 2004). The Florida State Supreme Court, for example, used the fact that private school curricula are not "uniform" as the basis for declaring an early state voucher program unconstitutional (*Bush v. Holmes*, 2006). Finally, critics argue that private schools may be effective at teaching advantaged students but are ineffective at serving the kinds of disadvantaged students who participate in school voucher programs (Hill & Scott, 2017).

The extent to which students benefit from vouchers, or not, is an empirical question (Doolittle & Connors, 2001). In this article, we estimate the causal impact of the Louisiana Scholarship Program (LSP), a statewide publicly financed voucher program, on student achievement for the first cohort of program applicants four years after the program was implemented. We determine the program's impacts in English Language Arts (ELA), math, and science, as well as its impact on college enrollment. Originally piloted in New Orleans in 2008 as the Student Scholarship for Educational Excellence Program, the LSP expanded statewide in the school year 2012–13, allowing almost 5,000 low- to moderate-income students across Louisiana to transfer out of low-performing public schools and into private schools.

Our analyses use oversubscription lotteries to generate unbiased estimates of the participant impacts of the LSP.[1] We use admission lotteries for student's first-choice, or top-ranked,[2] private school as instrumental variables to estimate the causal effect of using an LSP scholarship to enroll in a participating private school for applicants induced to attend that school as a result of winning the lottery (Bloom & Unterman, 2014). Our analysis uses student-level data obtained via a data-sharing agreement with the state of Louisiana along with college enrollment data from the National Student

[1]We describe our research design as "lottery-based" and "causal." We do not call it "experimental" because researchers disagree regarding whether or not treatment-on-the-treated estimates using lotteries as instrumental variables, such as the ones featured in our study, are truly "experimental." We do not enter that debate here. We characterize our findings as "causal" because there is general agreement that random lotteries are the ideal instrumental variables and that sound instrumental variables recover unbiased causal estimates of the impact of a treatment on the self-selected complier subgroup of participants offered the treatment (Angrist et al., 1996; Bloom & Unterman, 2014; Murray, 2006).

[2]Eligible LSP applicants could submit up to five rank-ordered private school preferences. We focus on first-choice school lotteries to ensure independence of treatment assignment, as whether or not a student won a lottery for placement in a lower-choice school likely was influenced by factors such as the number and popularity of non-first-choice schools listed which could bias comparisons of "any-lottery winners" to "no-lottery winners."

Clearinghouse. We operationalize achievement as student performance on the criterion-referenced tests in English Language Arts, math, and science that the state mandates for public school accountability purposes. College entrance is a dichotomous measure defined if students enrolled, or not, in at least one semester within six months of their expected high school graduation.

Previous research indicates large negative impacts of LSP voucher usage on ELA, math, and science achievement after one year of participation (Abdulkadiroğlu et al., 2018; Mills, 2015) that appear to attenuate over the following two years (Mills & Wolf, 2017a, 2017b). Our study indicates that the trend of improvement in LSP scholarship usage effects observed in prior research did not continue into the fourth year of the program. Although the negative effects students experienced in the first year of the program are substantially reduced by year four, students who used LSP vouchers to enroll in LSP participating private schools at any point between 2012-13 and 2015-16 continue to score significantly worse on state assessments than their control group counterparts. These effect estimates are statistically significant in most models and are negative in all tested subjects. We observe little evidence of differentiation in achievement effects by gender. In contrast, African American students experience significantly less negative voucher usage impacts relative to other students.

Using the same lottery-based analytical strategy, we also estimate the effect of LSP usage on the likelihood that students enter college. We estimate that 42.0% of scholarship users entered any college compared to 38.8% of their control group counterparts; however, this difference of 3.2 percentage points is not statistically distinguishable from zero. In addition, we find no significant difference in the rates that scholarship users and their control counterparts enter two- or four-year institutions.

Our research contributes to the existing literature on the participant effects of school voucher programs in multiple ways. First, it uses a highly rigorous lottery-based design to estimate causal programmatic effects by avoiding self-selection bias concerns. Second, ours is just the second lottery-based evaluation of a private school voucher program in the U.S. that includes both achievement and attainment results.[3] Third, it is among the first evaluations of the new wave of statewide school voucher programs (Abdulkadiroğlu et al., 2018; Figlio & Karbownik, 2016; Mills & Wolf, 2017a; Waddington & Berends, 2018) which are replacing the more heavily studied urban initiatives of the past. Finally, this study examines medium-run student outcomes of achievement effects in ELA, math, and science after four years, as well as initial educational attainment effects. These outcomes, taken jointly, provide a more thorough understanding of how the effects of this voucher program evolved over time than is the case for most other voucher evaluations.

This paper proceeds as follows. In the next section, we provide a brief background on vouchers as a policy instrument in K-12 education and summarize the evidence of their effects on student achievement and attainment. We then describe the LSP and the lottery process that enabled our causal analysis. Next, we discuss the data and analytical strategy used to estimate the participant effects. We then present the results of our analyses and discuss our findings.

[3]Wolf et al. (2013) also used a lottery-based design estimating the achievement and high school graduation effects of the D.C. Opportunity Scholarship Program.

School Vouchers and K-12 Education

School voucher initiatives tend to be justified based on some mix of economic and social justice criteria. Friedman (1955) famously argued that government should fund K-12 public education due to the positive spillovers educated citizens create for the broader society. However, he additionally claimed government-run schools need not be the sole or even the main delivery mechanism for publicly funded education. The traditional public-school system unnecessarily operates as a monopoly, Friedman observed, with the flaws endemic to monopolies including inefficiency, ineffectiveness, and unresponsiveness to their captive audience of customers.

Other scholars emphasize social justice as a motivation for vouchers (Viteritti et al., 2005). John Coons (forthcoming) observes that families of wealth buy their children's way into desirable schools by purchasing expensive homes in upscale public school districts, a practice called "Tiebout school choice" (Tiebout, 1956)or by self-financing private schooling. "The rich buy autonomy," Coons writes, "the rest get conscripted" into low-performing government-run schools. "'Public'? To the contrary, the system is a Balkanized plutocracy." (p. 118). Brighouse (2000) points out while discussing the U.S. system of residential assignment to government-run schools, "There is something deeply inequitable about a system that effectively accords parental choice only to the wealthy." (p. 32)

The type and extent of government regulations on voucher programs tend to distinguish more free-market-oriented programs, such as Florida's McKay Scholarships for Students with Disabilities Program, from more social-justice-oriented programs, such as the Milwaukee Parental Choice Program. The Florida program, which supported almost 29,000 students in the fall of 2019, is open to students with disabilities of all income levels (EdChoice, 2020, 31–32). Florida private schools participating in the program can apply admissions standards to McKay Scholarship students and are not required to administer an achievement test for accountability purposes. The Milwaukee program, in contrast, which similarly enrolled around 29,000 students in 2019, is limited to low-income students, requires that participating private schools admit students by random lottery when over-subscribed, and mandates the administration of the state criterion-referenced test to all voucher students in grades tested by the public schools (EdChoice, 2020, 77–78). These three regulations: income requirements, lottery admissions, and state testing, distinguish the more social-justice-focused voucher programs from their more free-market-focused cousins.

Regardless of policy design, sound research design is critical when evaluating school voucher programs (Doolittle & Connors, 2001; Jung & Pirog, 2014). Parents who seek educational alternatives for their children likely differ from non-choosing parents in unmeasurable ways that might affect student outcomes, generating possible selection bias (Murnane, 2005). Fortunately, many studies examining school vouchers in the United States have used lottery-based or matching research designs that attempt to control for unmeasured factors. In the sections that follow, we summarize findings from studies of U.S. voucher programs using either gold-standard lottery-based or rigorous quasi-experimental research designs.

Table 1. Participant achievement effects of school voucher programs in the U.S. on reading test scores, by geographic scope of program.

Lead author	Study type	Location	Study year	Outcome year	Overall findings
Statewide					
Egalite	QED	North Carolina	2020	1	+0.44 standard deviations
Waddington	QED	Indiana	2018	4	Null
Abdulkadiroglu	Lottery-based	Louisiana	2018	1	−0.08 standard deviations
Mills	Lottery-based	Louisiana	2017	3	Null
Figlio	QED	Ohio	2016	3	−0.31 standard deviations
Urban only					
Webber	Lottery-based	D.C.	2019	3	Null
Anderson	Lottery-based	D.C.	2017	4	+0.24 standard deviations
Wolf	Lottery-based	D.C.	2013	4	+0.13 standard deviations
Witte	QED	Milwaukee	2012	4	+0.15 standard deviations
Metcalf	QED	Cleveland	2003	5	Null
Greene	Lottery-based	Milwaukee	1998	4	+0.26 standard deviations
Peterson	QED	Cleveland	1998	1	+0.16 standard deviations
Rouse	Lottery-based	Milwaukee	1998	4	Null
Witte	QED	Milwaukee	1998	4	Null

Notes: "Study Year" refers to the year of publication. QED stands for quasi-experimental design. "Outcome Year" is the final year post-baseline in which achievement effects were reported. A "null" finding is one that was not statistically significant at or beyond $p < 0.10$. Overall findings are taken from the authors' preferred statistical model. Findings that are positive and statistically significant appear with no shading while those that are null appear with light shading and those that are negative and statistically significant appear with dark shading. Findings from lottery-based studies are the Impact-on-Treated (IOT) estimates when available, otherwise they are the Intent-to-Treat estimates. *Sources*: Abdulkadiroğlu et al. (2018); Anderson and Wolf (2017); Egalite et al. (2020); Figlio and Karbownik (2016); Greene et al. (1998); Metcalf et al. (2003); Mills and Wolf (2017b); Peterson et al. (1998); Rouse (1998); Waddington and Berends (2018); Webber et al. (2019); Witte (1998); Witte et al. (2012); Wolf et al. (2013).

Literature on Private School Vouchers and Student Achievement

The effects of school vouchers on participants' achievement remain contested (Wolf et al., 2018). We are aware of 14 previous U.S.-based studies that applied lottery-based or quasi-experimental methods to data from voucher programs in Cleveland, Ohio; the District of Columbia; Indiana; Louisiana; Milwaukee, Wisconsin; North Carolina; and Ohio to determine their impacts on student achievement.[4] Table 1 summarizes the estimated effects of these voucher programs on student reading or ELA test scores.[5] The spread of results from the authors' preferred analytic estimates for the final year of those evaluations regarding the reading effects of voucher programs is mixed but tilts positive. Six studies report that vouchers have a positive effect on reading test scores (Anderson & Wolf, 2017; Egalite et al., 2020; Greene et al., 1998; Peterson et al., 1998; Witte et al., 2012; Wolf et al., 2013). Two evaluations conclude that the reading effects of vouchers are negative (Abdulkadiroğlu et al., 2018; Figlio & Karbownik, 2016). The remaining six studies are inconclusive regarding the reading effects of vouchers

[4]We know of 12 additional evaluations of the achievement effects of tax-credit or privately funded K-12 scholarship programs that some commentators include in the school voucher literature because, like vouchers, those programs increase student access to private schools of choice. We exclude tax-credit and privately funded scholarship studies from this review because those initiatives differ from voucher programs in three key ways: (1) they are privately, not publicly, funded, (2) they tend to be much more lightly regulated than voucher programs, and (3) the values of the scholarships tend to be much lower than the values of vouchers. Were we to add the tax-credit and privately funded K-12 scholarship studies to this review, our basic generalizations about the literature would not substantially change.

[5]We use $p < 0.10$ as the minimum threshold for statistical significance because some of the early voucher evaluations drew on small samples and therefore used that marginal level of significance to reduce the risk of Type II errors (Greene et al., 1998; Peterson et al., 1998; Rouse, 1998; Witte, 1998). We also report reading and ELA scores in the same table. Some studies used ELA and others used reading test scores. For simplicity, we refer to all as reading effects.

(Metcalf et al., 2003; Mills & Wolf, 2017b; Rouse, 1998; Waddington & Berends, 2018; Webber et al., 2019; Witte, 1998). The pattern of reading effects of school vouchers is identical between the subgroups of lottery-based (3 positive, 1 negative, 3 null) and quasi-experimental (3 positive, 1 negative, 3 null) studies. The reading effects of vouchers from statewide studies (1 positive, 2 negative, 2 null) are both less common and less positive than the results from city-focused evaluations (5 positive, 0 negative, 4 null).

The math effects of vouchers are more thoroughly mixed than the reading effects (Table 2). Of the 14 studies, only four report positive effects of school vouchers on math test scores (Egalite et al., 2020; Greene et al., 1998; Peterson et al., 1998; Rouse, 1998). Three prior evaluations conclude that the math effects of vouchers are negative (Abdulkadiroğlu et al., 2018; Figlio & Karbownik, 2016; Waddington & Berends, 2018). The remaining seven studies of the math effects of vouchers are inconclusive (Anderson & Wolf, 2017; Metcalf et al., 2003; Mills & Wolf, 2017b; Webber et al., 2019; Witte, 1998; Witte et al., 2012; Wolf et al., 2013). The findings on the math effects of school vouchers are similar between the subgroups of lottery-based (2 positive, 1 negative, 4 null) and quasi-experimental (2 positive, 2 negative, 3 null) studies. The math effects of vouchers from statewide studies (1 positive, 3 negative, 1 null) are decidedly more negative than the results from city-focused evaluations (3 positive, 0 negative, 6 null).

We can only speculate as to why the evaluations of the achievement effects of voucher programs in cities tend to yield more positive (less negative) findings than the studies of statewide voucher initiatives. One possible reason is the type of test used in the studies. Except for the Witte et al. (2012) study of Milwaukee, the other city-focused evaluations all use norm-referenced tests to gauge the achievement effects of the programs. The only statewide voucher evaluation to use a norm-referenced test, by Egalite and her colleagues (2020) of the North Carolina voucher program, reports large positive achievement effects. Three of the four statewide studies of vouchers rely on official state criterion-referenced tests. Those evaluations report negative voucher effects either just on math test scores (Waddington & Berends, 2018) or on both reading and math outcomes (Abdulkadiroğlu et al., 2018; Figlio & Karbownik, 2016).[6] It is possible, perhaps even likely, that the state criterion-referenced tests are less closely aligned to the curricula used in the private schools serving voucher students than they are to the official state curriculum mandated in the public schools serving the comparison students in the studies. Program scale and the extent of customer choice also could be factors, as the city-based programs demonstrating more positive achievement effects are smaller and tend to offer students multiple private schooling options within commuting distance of their homes while the statewide programs tend to be larger in scale and serve at least some students in rural areas without many nearby private schools.

Voucher program test score effects often vary across the outcome years of longitudinal evaluations. The original evaluation of the District of Columbia Opportunity Scholarship Program (OSP) reports significant positive impacts in reading after three years (Wolf et al., 2009, p. 36) that are only significant at a 94 percent level of

[6]As we explain in a later section, the LSP evaluation relied on a new criterion-referenced test in Year 3 of the evaluation, and both public and private schools in Louisiana were exempt from accountability sanctions that first year, which might explain why no significant voucher effects on achievement were identified in Mills & Wolf, 2017b.

Table 2. Participant achievement effects of school choice programs in the U.S. on math test scores, by geographic scope of program.

Lead author	Study type	Location	Study year	Outcome year	Overall findings
Statewide					
Egalite	QED	North Carolina	2018	1	+0.36 standard deviations
Waddington	QED	Indiana	2018	4	–0.15 standard deviations
Abdulkadiroglu	Lottery-based	Louisiana	2018	1	–0.41 standard deviations
Mills	Lottery-based	Louisiana	2017	3	Null
Figlio	QED	Ohio	2016	3	–0.54 standard deviations
Urban only					
Weber	Lottery-based	D.C.	2019	3	Null
Anderson	Lottery-based	D.C.	2017	4	Null
Wolf	Lottery-based	D.C.	2013	4	Null
Witte	QED	Milwaukee	2012	4	Null
Metcalf	QED	Cleveland	2003	5	Null
Greene	Lottery-based	Milwaukee	1998	4	+0.40 standard deviations
Peterson	QED	Cleveland	1998	1	+0.31 standard deviations
Rouse	Lottery-based	Milwaukee	1998	4	+0.23 standard deviations
Witte	QED	Milwaukee	1998	4	Null

Notes: "Study Year" refers to the year of publication. QED stands for quasi-experimental design. "Outcome Year" is the final year post-baseline in which achievement effects were reported. A "null" finding is one that was not statistically significant at or beyond $p < 0.10$. Overall findings are taken from the authors' preferred statistical model. Findings that are positive and statistically significant appear with no shading while those that are null appear with light shading and those that are negative and statistically significant appear with dark shading. Findings from lottery-based studies are the Impact-on-Treated (IOT) estimates when available, otherwise they are the Intent-to-Treat estimates. *Sources*: Abdulkadiroğlu et al. (2018); Anderson and Wolf (2017); Egalite et al. (2020); Figlio and Karbownik (2016); Greene et al. (1998); Metcalf et al. (2003); Mills and Wolf (2017b); Peterson et al. (1998); Rouse (1998); Waddington and Berends (2018); Webber et al. (2019); Witte (1998); Witte et al. (2012); Wolf et al. (2013).

confidence in the fourth year of the study (Wolf et al., 2013). The recently completed lottery-based study of the OSP finds that first- and second-year negative program impacts in math fade out by the third year of the analysis (Webber et al., 2019). An evaluation of the Milwaukee voucher program concludes that a combination of the choice program and a high-stakes testing policy generate test score gains in reading, but not math, and only in the study's fourth year (Witte et al., 2014). The initial achievement effects reported for school voucher program participants after one or two years often change when those same effects are evaluated after three or four years, possibly because students receive a stronger "dose" of the voucher treatment over multiple years but also perhaps because the samples of tested students change over time.

School voucher studies have yet to identify consistent moderators of achievement effects. During the first few years of the Indiana evaluation (Waddington & Berends, 2018) and the Louisiana Scholarship Program evaluations (Tuchman & Wolf, 2017), students with disabilities appear to have escaped the negative achievement effects of those voucher programs that were evident for their peers without disabilities. Waddington and Berends (2018) further report that Black and Latinx students experience significantly more positive (less negative) achievement effects from the Indiana Scholarship Program than do white participants.[7] Wolf and his colleagues (2013) find that students with higher previous performance, students applying from public schools not classified

[7]Like Waddington and Berends (2018), Howell and his colleagues (2002) report that Black students realize more positive achievement effects from private schools of choice than do their white and Latinx peers. Those results, however, are from evaluations of three privately-funded K-12 scholarship programs with different program designs and eligibility criteria than most school voucher programs. Thus, we do not highlight those findings here.

as "in need of improvement," and female students disproportionately benefit from voucher receipt in the D.C. OSP. Witte and his colleagues (2012), in contrast, report that the achievement effects of the Milwaukee Parental Choice Program appear to be more positive for students with lower previous performance when examining reading test scores but more positive for students with higher previous performance when examining math outcomes.

The school voucher literature contains even less consensus on the possible mediators of voucher achievement effects. Waddington and Berends (2018) report that the initial negative achievement effects of the Indiana program were significantly lower for the subgroup of students who attended Catholic schools compared to those who attended non-Catholic private schools. Lee and his colleagues (2020) find at least suggestive evidence that the opposite may have been true in the first two years of the Louisiana program. They also report that voucher students who attended private schools with larger enrollments, more school personnel per grade, a library, and located in a city experience more positive (less negative) voucher effects on math scores than voucher students who attended private schools without those features. None of their findings, however, survive all their robustness tests.

The pattern of results from previous lottery-based and quasi-experimental evaluations of voucher programs reflects noticeable effect heterogeneity, with estimated effects across studies ranging from negative to positive, differing somewhat in reading and math, varying across outcome years, and especially differing between city-focused evaluations and studies of broader-scoped and larger-scale statewide programs. No previous study has examined the effect of a U.S. school voucher program on student science scores.

Literature on Private School Choice and Student Attainment

The literature is much smaller on the effects of private school vouchers on students' educational attainment as measured by high school graduation, college entrance, and degree completion. This research base is less developed than the school voucher achievement impacts literature because attainment evaluations require following students for many years after their initial experience in the program. Educational attainment is a more direct proxy for student success than student test scores and is strongly associated with a host of positive long-term life outcomes. Higher levels of educational attainment are predictive of a longer, healthier, and more economically productive life (Belfield & Levin, 2007; Day & Newburger, 2002; Meara et al., 2008; Muennig, 2008; Muenning, 2005). Moreover, the shorter-term achievement effects of public and private school choice programs seldom predict the same program's longer-term attainment effects. Some private and public school choice programs demonstrate large positive test score impacts for students but null or negative effects on their post-secondary outcomes; while other such programs show no effect on test scores but large positive effects on attainment (Hitt et al., 2018). Examining both achievement and attainment outcomes provides a more comprehensive understanding of the LSP's impact on students' future success.

Only five studies assess the impact of private school vouchers on student attainment in just two programs (Table 3): the Milwaukee Parental Choice Program (MPCP) and

Table 3. Participant attainment effects of school voucher programs in the U.S.

Lead author	Study type	Location	Study year	Outcome	Overall findings
Wolf	QED	Milwaukee	2019	B.A.	Null or +3 percentage points
Wolf	QED	Milwaukee	2019	Enrolled	+4–6 percentage points
Chingos	Lottery-based	D.C.	2018	Enrolled	Null
Cowen	QED	Milwaukee	2013	Enrolled	+4–5 percentage points
Wolf	QED	Milwaukee	2019	Diploma	+7 percentage points
Cowen	QED	Milwaukee	2013	Diploma	+2–7 percentage points
Wolf	Lottery-based	D.C.	2013	Diploma	+21 percentage points
Warren	QED	Milwaukee	2011	Diploma	+12 percentage points

Notes: "Study Year" refers to the year of publication; QED stands for quasi-experimental design. A "null" finding is one that was not statistically significant at $p < 0.10$ or better. "B.A." designates the outcome of the award of a Bachelor's degree from a 4-year college or university. "Enrolled" designates the outcome of ever being enrolled in a 4-year college or university within the timeframe of the evaluation. "Diploma" designates the outcome of graduating from high school. Overall findings are taken from the authors' preferred statistical model. Findings that are positive and statistically significant appear with no shading while those that are null appear with light shading and those that are negative and statistically significant appear with dark shading. *Sources*: Chingos (2018); Cowen et al. (2013); Warren (2011); Wolf et al. (2013); Wolf et al. (2019).

the District of Columbia Opportunity Scholarship Program (D.C. OSP). Two studies consider high school graduation only (Warren, 2011; Wolf et al., 2013), one study considers college enrollment only (Chingos, 2018), and two studies examine both (Cowen et al., 2013; Wolf et al., 2019).

Of the four total studies that consider the effect of private school choice on the likelihood of students graduating from high school, all report statistically significant positive effects. The largest impact is in the lottery-based D.C. OSP evaluation, where voucher usage is estimated to increase the likelihood of graduating from high school by 21 percentage points (Wolf et al., 2013). Using student matching methods, Cowen et al. (2013) find that students participating in the MPCP are two to seven percentage points more likely to graduate from high school in four years compared to similar peers in traditional public schools, an initial finding largely replicated by a follow-up study with different student samples (Wolf et al., 2019). Evaluating the same program, but with program-level data, Warren (2011) finds that voucher students are twelve percentage points more likely to graduate in six years compared to the state average high school graduation rate.

The voucher impacts on college enrollment and degree attainment appear to be more varied than on high school graduation. The two Milwaukee studies both find that students in the voucher program are four to six percentage points more likely to enter four-year colleges and persist in them longer than matched public school students in most estimations (Cowen et al., 2013; Wolf et al., 2019). In contrast, Chingos (2018) reports that students in the D.C. program do not exhibit significant college enrollment benefits from the initiative. Finally, Wolf and his colleagues (2019) find that the older cohort of students in their Milwaukee study obtained bachelor's degrees at similar rates regardless of whether they were in the voucher program or the comparison group. Voucher students in the younger cohort of that study demonstrated a three percentage point advantage in their rate of earning a four-year college degree compared to students in the matched comparison group, which represents an increase of nearly 40 percent in the likelihood of degree attainment.

Overall, private school voucher programs tend to have a significant positive effect on students' likelihood of graduating from high school and a neutral to positive effect on enrolling in and graduating from postsecondary institutions. However, research remains limited. Five studies have considered the attainment effects of only two voucher programs in the U.S. Moreover, only two of those evaluations employed a gold standard, lottery-based design. Our study expands the nascent literature on the effects of voucher programs on postsecondary outcomes by evaluating the causal impact of the Louisiana Scholarship Program on students' likelihood of entering college in addition to ELA, math, and science achievement outcomes.

Description of the Intervention

The Louisiana Scholarship Program (LSP) is a statewide school voucher initiative available to moderate- to low-income students attending low-performing public schools. The program is limited to students who satisfy both of these criteria: (1) they have family incomes at or below 250 percent of the federal poverty line; and (2) they attend a public school that was graded C, D, or F for the prior school year according to the state's school accountability system, are entering kindergarten, or are enrolled in the Recovery School District.[8] In the initiative's first year, 9,736 students applied. Students in our analytic samples are located throughout the state with most of them living in large urban areas such as New Orleans, Shreveport, and Baton Rouge. Applicants are disproportionately entering the lower grades: 55 percent for kindergarten through the third grades, 26 percent for the fourth through the sixth grades, and 19 percent for the seventh through the twelfth grades.

The state-allocated voucher amount is the lesser of the student's public school state and local funding or the tuition charged by the participating private school the student attends. In the 2012–13 school year—the year in which our analysis cohort first applied for LSP vouchers—average tuition at participating private schools ranged from $2,966 to $8,999, with a median cost of $4,925, compared to an average total minimum foundation program per-pupil funding of $8,500 for Louisiana public schools that same year.

To participate in the scholarship program, private schools must meet certain state government regulations involving admissions, financial practices, student mobility, and the health and safety of students. Surveys of participating and nonparticipating private schools in Louisiana suggest that the program's regulatory requirements influenced private schools' decisions to participate or not (Kisida et al., 2015; Sude et al., 2018), potentially explaining why only a third of Louisiana private schools opted into the program in 2012–13.[9]

Research Methodology

Lottery-Based Causal Design

We exploit lotteries in oversubscribed private schools to estimate the causal effect of the LSP on students' achievement as well as their likelihood of entering college.

[8]In 2012-13, the Recovery School District included most of the public schools in the city of New Orleans, several in Baton Rouge, and a single school in Shreveport, Louisiana.

[9]This low participation rate could also be due to the existence of other private school choice programs in the state. Louisiana also provides parents who self-finance their child's private schooling with an annual state tax deduction of up to $5,000 per child, an indirect benefit for private schools in the state for which they face no additional state regulations.

Lottery-based designs are widely viewed as the gold standard for evaluation because they hold great potential to identify causal effects (Mosteller & Boruch, 2002; Pirog et al., 2009; Rossi et al., 2004). Our analytic strategy mitigates the risk of selection bias, allowing us to estimate the causal effect of participating in the LSP.

Eligible students apply for an LSP scholarship through a centralized enrollment process that the Louisiana Department of Education (LDOE) administers. When applying, families can rank order their top five preferred private schools. The allocation algorithm prevents gaming, incentivizing families to reveal their true school preference rankings. This process is similar to the deferred acceptance lottery used in New York City to assign students to schools through the city's public school choice program (Abdulkadiroğlu et al., 2005).

The LSP enrollment system awards scholarships based on student priority status and seat availability in preferred private schools. First, students with disabilities and "multiple birth siblings," such as twins and triplets, are automatically awarded LSP scholarships, if space exists at their preferred school. The remaining students are assigned one of six priorities:

- **Priority 1** – Students who received LSP scholarships in the prior school year who are applying to the same school
- **Priority 2** – Non-multiple birth siblings of Priority 1 awardees in the current round
- **Priority 3** – Students who received LSP scholarships in the prior school year who are applying to a different school
- **Priority 4** – New applicants who attended public schools that received a "D" or "F" grade in Louisiana's school accountability system at baseline
- **Priority 5** – New applicants who attended public schools that received a "C" grade
- **Priority 6** – New applicants who are applying to kindergarten

The process begins by attempting to place all Priority 1 students into their first-choice school by grouping all students applying to the same school and grade combination, and then by checking the number of available seats for that grouping (Figure 1). If there are more seats than applicants, all students receive a scholarship. If there are no seats available, no student receives a scholarship. If there are more applicants than seats, students are awarded scholarships through a lottery. Once the process is complete for Priority 1 students, the algorithm attempts to place Priority 2 students into their first-choice school using the same decision rules. After cycling through all remaining priority categories, the LSP algorithm moves to a second stage of the allocation process and tries to place the remaining students into their second-choice schools. The process continues until all eligible applicants have or have not been awarded a scholarship.

Only a subset of eligible applicants participated in a lottery: students in Priority groups 1 through 6 whose school-grade combination had more applicants than seats. Using data on student characteristics and school preferences, we identify a lottery as occurring when the percentage of students awarded an LSP scholarship falls between 0 and 100 percent for a given combination of priority category, school, and grade. We focus on this subset of LSP applicants facing lotteries for their first-choice school to

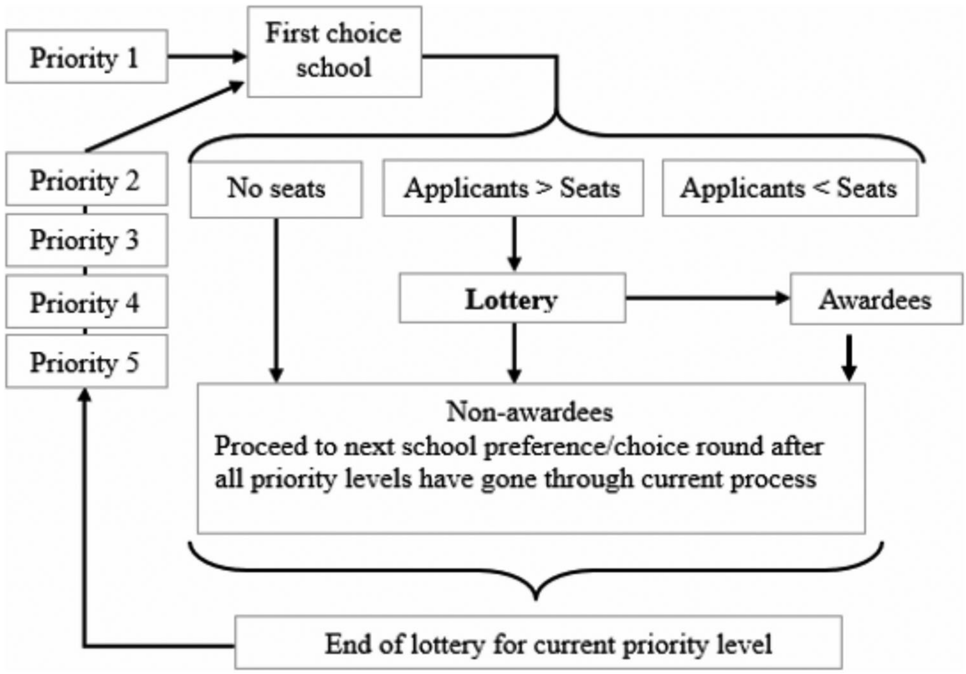

Figure 1. First Stage of the Louisiana Scholarship Program (LSP) award allocation process for the 2012–2013 school year. *Notes:* This figure illustrates the iterative process used to allocate LSP scholarships to students. In addition, this figure highlights the fact that only a subset of students was awarded LSP scholarships via lotteries. Our analysis focuses on isolating lotteries for one's first-choice school. Source: Mills and Wolf (2017a).

estimate the causal effects of the LSP on participants' achievement and the likelihood of entering college. The focus on first-choice school lotteries ensures that an individual's own scholarship assignment is independent of other student lottery outcomes. First-choice school lotteries have been used to study the relationship between public school choice and post-secondary outcomes (Deming et al., 2014) as well as the effects of small high schools on student achievement (Bloom & Unterman, 2014).

The schools in our sample are more likely to be popular ones, as over-subscription lotteries can only occur at schools where there are more applicants than seats available.[10] Moreover, higher-quality schools are often more likely to be oversubscribed than lower-quality schools (Abdulkadiroglu et al., 2011). These points suggest that the estimates presented here are likely the upper bounds of the program's average effect on student achievement and attainment.

Data Description

The data for this study come from two sources. The LDOE provided us with student program application information, administrative records, and student performance on the Louisiana state assessments. We additionally obtained college enrollment data from the

[10]Refer to Appendix C for an analysis predicting the likelihood of a student being in a first-choice lottery as a function of student characteristics.

National Student Clearinghouse (NSC) for LSP applicants who applied for grades seven through twelve in the first year of the program.

Achievement Analysis Data

Our analysis of the LSP's impact on student achievement uses LDOE-provided student application and administrative records. These data include information on students' school choice sets, results of placement lotteries, demographic background, and achievement on Louisiana's statewide assessments.

The Louisiana state accountability system emphasizes test-based accountability. We use student performance on the Louisiana state assessments in grades three through eight as our outcome measure of interest in the achievement analysis. Private schools are required to test all students participating in the LSP using the state accountability assessment for any grade in which the public school system also tests its students. The 2011–12 (baseline), 2012–13 (Year 1 Outcome), and 2013–14 (Year 2 Outcome) assessment data contain student scores on the LEAP and iLEAP exams, criterion-referenced tests aligned to Louisiana state education standards. The 2014–15 (Year 3 Outcome) data instead provide student scores in ELA and math on the PARCC, a criterion-referenced test aligned with the Common Core standards. In science, in contrast, students continued to take the LEAP/iLEAP exams aligned with state standards.[11] Beginning in 2015–16 (Year 4 Outcome), Louisiana students took the LEAP 2025 assessments, a revamped version of the LEAP/iLEAP, in grades three through eight in ELA, math, and science. These assessments are aligned with the 2014–15 PARCC tests (Louisiana Department of Education, 2015).

College Entrance Analysis Data

In addition to the application data the LDOE provide, we use data from the National Student Clearinghouse (NSC) Student Tracker Service for the college entrance analysis. The NSC collects data on college entrance, persistence, and degree attainment from nearly all public and private post-secondary institutions in the U.S. The comprehensiveness of the NSC database allows us to capture records for students in our sample who attend college within or outside of Louisiana. We focus on college enrollment, in both two- and four-year institutions, as few students in our sample are old enough to have completed a degree. Seventh-grade LSP applicants, in the first year of the program, could have enrolled in one semester of college by fall 2018 and twelfth grade applicants could have enrolled in up to five and a half years of college by fall 2018. We consider students "entering college" if they enrolled in at least one semester of courses within six months following their expected high school graduation, as confirmed by the NSC data.[12] We believe this is a conservative estimate of college entrance as many students

[11]PARCC assessments were administered as paper and pencil tests in grades three through eight for both ELA and math. The spring 2015 administration of PARCC assessments was considered a transition period by LDOE, with no summer retest period available. Student performance on PARCC assessments factored into performance scores for both public and private schools; however, the schools were graded on a curve in the 2014–15 school year.

[12]We had planned to examine the impact of the LSP on high school graduation rates, but the graduation database used a new anonymous student ID that could not be linked to our student database.

enter college a year or two following high school graduation; however, we only observe the seventh-grade applicants six months past their expected high school graduation.

Analytical Strategy

Local Average Treatment Effect Estimation

The LSP scholarships are awarded through a deferred acceptance algorithm, complicating our attempt to identify the program's impact on student outcomes because assignments to lower-ranked schools depend on the outcomes of earlier lotteries. We still leverage the random assignment of first-choice school lotteries to estimate the program's causal effect; however, doing so limits the policy relevance of the traditional intent-to-treat (ITT) estimator as students can participate in multiple lotteries which, collectively, "intend" to treat all students (Bloom & Unterman, 2014). Instead of estimating the ITT experimental effect, we estimate the impact of LSP scholarship usage on student achievement and college entrance—also known as the Local Average Treatment Effect (LATE) (Angrist & Pischke, 2009; Cowen, 2008). Our LATE effects are generated by using the result of one's first-choice school lottery to instrument for scholarship usage at any participating private school in a 2-Staged Least Squares (2SLS) framework. This estimation method accounts for the fact that some first-choice lottery losers could win a later lottery and enroll in a private school. The 2SLS procedure effectively treats these students as control-group crossovers.[13] The result is an unbiased estimate of the effect of using an LSP scholarship to attend a participating private school for those who both faced and complied with their first-choice lottery assignment (Bloom & Unterman, 2014).

The lottery is an ideal instrumental variable as the high placement take-up rate for this program ensures that it is a strong predictor of private schooling while the random nature of the lottery process assures that scholarship receipt is uncorrelated with unobserved factors related to our outcomes (Murray, 2006). Because the lottery is the only way students can receive LSP scholarships to attend a participating private school, we are confident that the variable only influences student outcomes through the private schooling that it enables.

We use the following 2SLS model in our analyses, where i denotes student and j denotes lottery:

$$E_i = \sum \pi_j R_{ji} + \delta T_i + X_i \beta + u_i \tag{1}$$

$$Y_i = \sum \alpha_j R_{ji} + \tau \widehat{E}_i + X_i \gamma + \epsilon_i \tag{2}$$

- E_i indicates if a student used an LSP scholarship to enroll in any participating private school at any point between the 2012–13 and 2015–16 school years.[14]

[13]See Appendix B for details on treatment and control group school enrollment patterns.

[14]Prior evaluations of school voucher programs have examined enrollment effects in several ways. For example, Mayer et al. (2002) define enrollment as being "consistently enrolled in a private school," while Rouse (1998) defines enrollment as the number of years enrolled, in an attempt to capture potential dosage effects. By defining enrollment conservatively as "ever attending a private school," our study falls in line with Wolf et al. (2013).

- R_i is a fixed effect for a student's first-choice school lottery, which compares students within the same school-grade-priority category combination.[15]
- T_i indicates if a student received an LSP scholarship to his or her first-choice school.
- Y_i is the given outcome of interest. In the achievement analysis, it represents standardized student ELA, math, or science test scores in Year 4 of the program (2015–16).[16] In the college entrance analysis, it represents whether or not a student enrolled in at least one semester of college within six months following expected high school graduation (coded 1) or not (coded 0).[17]
- X_i is a vector of student characteristics collected at baseline (2011–12) from the student's LSP application form.

The 2SLS procedure produces an unbiased LATE estimate ($\hat{\tau}$) for the program. We account for the nesting of students within lotteries using bootstrapped standard errors (Angrist & Pischke, 2009). The families of students and their post-treatment schools could represent additional nesting factors (Wolf et al., 2013). The results presented here do not account for these sources of nesting due to the complex nature of multi-level clustering. Clustering on lottery should capture a large amount of the nesting of individuals within current schools, as each lottery places students in a specific school. Moreover, siblings constitute only around 5 percent of our analytical samples in both the achievement and college entrance analysis and therefore are not a substantial nesting concern.

Sample Selection

For both the student achievement and college entrance analysis, our sample consists of eligible LSP applicants in the first year of the statewide program who also participated in over-subscription lotteries for their first-choice private school. We exclude students with special education designations on their applications as well as those who are multiple birth siblings since the program guarantees those students scholarship placements. The student-level data that the LDOE provides indicate an initial sample of 9,736 eligible LSP applicants in the first year of the program's statewide expansion.[18] Of these, 5,296 students received LSP scholarship placements in a specific private school and 4,440 did not receive a voucher-supported placement. The achievement and college entrance analyses rely on different samples of 2012–13 applicants, with limited overlap between the two samples (Figure 2).

[15]We include fixed effects for first-choice school lotteries to account for differing probabilities of success across lotteries (Gerber & Green, 2012). By using fixed effects, we are comparing lottery winners and losers within the same lottery strata to calculate unbiased estimates of the effect of being randomly offered an LSP scholarship. The approach is comparable to analyzing the impact of hundreds of "mini-experiments" and aggregating the results across them.

[16]Student achievement scores are standardized using distributional parameters of outcomes from the control group.

[17]College entrance effects are estimated using linear probability models due to the difficulty of a probit or logit model achieving convergence given the large number of lottery fixed effects in our models. For each of our models, the linear predictions fall within the appropriate zero to one range.

[18]LDOE provided us data only on the first cohort of student applicants and not for subsequent cohorts. The LSP continues today with new cohorts of students applying every year. The 2012–13 cohort has been the largest annual enrollment cohort, by far.

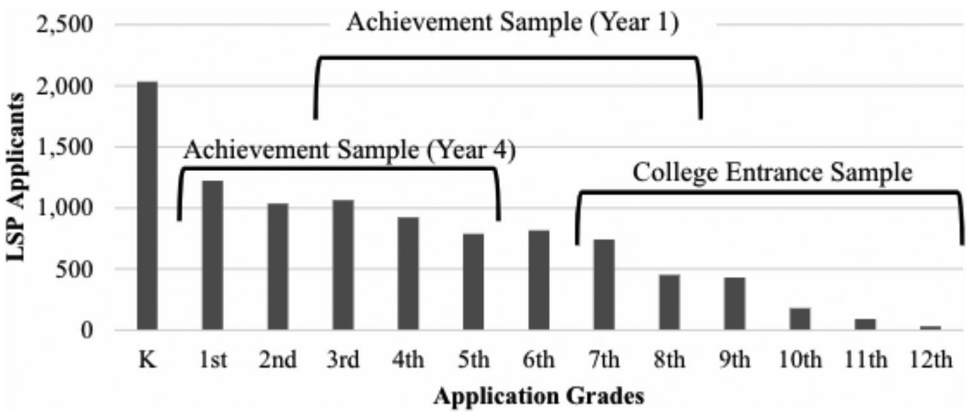

Figure 2. Achievement and college entrance samples by students' applicant grade. *Notes:* This figure illustrates the limited overlap of LSP applicants between the achievement and college entrance samples. The Achievement sample in the first year includes students who applied for the program in grades three through eight, where the college entrance sample includes students who applied for grades seven through twelve. Students who applied for kindergarten are not included in any of our analyses. *Source:* Authors' calculations.

Achievement Analysis Sample

Our primary achievement analysis sample includes students applying in the first year of the program for grades one through five. After excluding students with special education designations on their application as well as multiple birth siblings, there are 4,499 applicants for grades one through five. Of these students, 2,688 (60 percent) faced a lottery for their first-choice private school, 2,348 of whom also have outcome data in Year 4.[19] Of these students, 49 percent receive placement in their first-choice private school.[20]

College Entrance Analysis Sample

Our college entrance analytic sample focuses on the 1,927 students who applied for seventh through twelfth grades in the first year of the program. Of these students, 1,113 (58 percent) faced a lottery to gain admission to their first-choice private school.[21] Students who applied for a twelfth grade in Year 1 could have enrolled in up to five-and-a-half years of college by fall 2018, and students applying for seventh grade in Year 1 could have enrolled in one semester of college, assuming students graduated from high school

[19]Unlike in previous work (Mills, 2015; Mills & Wolf, 2017a; Mills & Wolf, 2017b), our primary analysis in this article does not include controls for baseline achievement. We do this for several reasons. First, requiring baseline test scores restricts our sample to only 780 students in grades three or four in school year 2011–12. A reduced sample size greatly decreases our power to detect a program effect. Second, while pre-test measures can increase the precision of our estimates, excluding baseline scores does not introduce any bias in a lottery-based analysis such as ours, assuming that the lotteries effectively randomized the treatment and control groups. Finally, requiring baseline scores also limits the generalizability of our findings beyond students applying for grades four and five. Appendix C examines the sensitivity of our results to the inclusion of baseline achievement data. Results are largely consistent with our primary analyses.

[20]Seventy-two percent of schools listed as first-choice schools by first- through fifth-grade applicants had at least one lottery in at least one of the given grades.

[21]Of the schools that seventh- through twelfth-grade applicants listed as their first-choice schools, 54 percent had at least one lottery in at least one of the given grades.

Table 4. Baseline equivalence of treatment and control groups on covariates.

	Achievement sample					College entrance sample				
	N (1)	Treatment mean (2)	Control mean (3)	Adjusted difference (4)	Se (5)	*N* (6)	Treatment mean (7)	Control mean (8)	Adjusted difference (9)	SE (10)
Female	2,348	0.54	0.51	0.01	0.03	1,113	0.50	0.53	−0.03	0.03
Race/ethnicity										
African–American	2,348	0.90	0.90	−0.03**	0.01	1,113	0.88	0.92	−0.03	0.02
Hispanic	2,348	0.02	0.02	0.01	0.01	1,113	0.03	0.03	0.00	0.03
White	2,348	0.05	0.05	0.01	0.01	1,113	0.06	0.04	0.02	0.02
Other	2,348	0.03	0.02	0.01	0.01	1,113	0.02	0.01	0.01	0.01
No. of school preferences	2,348	1.98	2.51	−0.13**	0.06	1,113	2.00	2.10	−0.10	0.06
Free-or reduced-price lunch	1,740	0.95	0.95	0.00	0.01	n/a	n/a	n/a	n/a	n/a
Standardized performance[a]										
ELA scale scores	780	−0.34	−0.34	−0.01	0.09	n/a	n/a	n/a	n/a	n/a
Math scale scores	779	−0.34	−0.34	−0.04	0.08	n/a	n/a	n/a	n/a	n/a
Science scale scores	779	−0.44	−0.44	0.07	0.07	n/a	n/a	n/a	n/a	n/a
Social studies scale score	779	−0.35	−0.34	−0.01	0.07	n/a	n/a	n/a	n/a	n/a

[a]Baseline test scores are only included for applicants in grades three through five. Scores are standardized within-grade based on the observed distribution of scale scores across Louisiana.

Notes: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. "Treatment" refers to students receiving LSP scholarships to their first-choice private school. All other students comprise the control group. "Adjusted Difference" is the difference between treatment and control group students, controlling for first-choice school lottery fixed effects. "SE" indicates the standard error of the difference, which accounts for clustering within lotteries. *Source*: Authors' calculations.

within four years. Given these differences between seventh- and twelfth-grade applicants, we only consider college enrollment within six months following a student's expected high school graduation regardless of the grade they were entering at the beginning of the study.

Baseline Equivalence

Lottery-based designs rely on randomization to create similar treatment and control groups. While we cannot know if members of the treatment and control groups differ on unobservable characteristics, we can get a good idea of the success of the first-choice lottery process by testing for equivalence in observable characteristics at baseline. Table 4 presents the results of this analysis as regression adjusted differences in means on key baseline covariates between the treatment and control groups for both the achievement and college entrance samples.

The results are favorable for our analysis. Nearly all the estimated adjusted differences between lottery winners and losers are statistically insignificant for both the achievement and college entrance samples (columns 4 and 9, respectively), suggesting that we have adequately identified random lotteries in our analytic samples. We observe that, on average, lottery winners in the achievement sample provided significantly fewer school preferences than lottery losers. We also observe a statistically significant difference between the treatment and control groups on the likelihood of being African American, even with comparisons made within-lottery. Given these differences, our preferred models include controls for the full set of variables we examined in Table 4. There are no significant observable differences between treatment and control students at baseline in the college entrance sample.

Table 5. Estimated effects of ever using an LSP voucher on student achievement after four years.

		Late			
	First stage (1)	Simple model (2)	+Test retake (3)	Fully specified (4)	Omitting new orleans (5)
English language arts	0.65***	−0.13*	−0.11	−0.22***	−0.21**
	(0.02)	(0.08)	(0.08)	(0.08)	(0.09)
Mathematics	0.65***	−0.32***	−0.30***	−0.39***	−0.39***
	(0.02)	(0.07)	(0.07)	(0.08)	(0.08)
Science	0.65***	−0.16*	−0.14	−0.21**	−0.23***
	(0.02)	(0.09)	(0.09)	(0.10)	(0.09)
Controls					
Test re-take			X	X	X
Demographics				X	X
No. school choices				X	
New orleans					
N	2182−2210		2182−2210	1606−1632	1405−1430
Lotteries	255		255	214−215	178−180

Notes. ***$p < 0.01$, **$p < 0.05$, *$p < 0.10$. *Simple model* refers to estimations that only control for first-choice school lottery fixed effects. *Test retake* indicates models including an indicator for if a student took the same subject test in 2 consecutive years. *Full model* refers to models controlling for test retaking, student demographics, the number of school preferences offered, and geography. Column (5) omits students who attended New Orleans public schools in 2011–12, to account for the existence of the New Orleans pilot program. Performance measures are standardized within-grade based on control group score distributions. All models include first-choice school lottery fixed effects. Standard errors (parentheses) account for clustering within lotteries. First stage F-statistics all exceed Staiger and Stock's (1997) recommended threshold of 10. *Source*. Authors' calculations.

Results

Student Achievement Results

We first present the results for our primary estimates of the impact of LSP scholarship usage on student achievement.[22] Prior work reported large declines in both math and ELA achievement after one year of voucher usage (Abdulkadiroğlu et al., 2018; Mills, 2015). These negative effects declined by half after Year 2 (Mills & Wolf, 2017a) and were not statistically significant after Year 3 (Mills & Wolf, 2017b). In contrast, the results presented here indicate large negative effects of ever using an LSP voucher by Year 4 on achievement on the Louisiana state assessments in ELA, math, and science. As with earlier work, the negative effect estimates are particularly large for math.

Table 5 reports our primary achievement effect estimates. Column 1 displays coefficient estimates for first-stage regressions. These coefficients identify the percentage of individuals complying with their initial lottery assignment, signaling that 65 percent of students in the analytical sample complied with their initial lottery assignment between 2012 and 13 and 2015 and 16.[23]

[22]Our primary analyses do not require baseline achievement, as this would restrict the sample to students applying for grades fouror five only. As a robustness check, we present results for models that additionally control for baseline achievement in Appendix C.

[23]We identify compliance as respecting the result of the first-choice lottery. Lottery compliers are treatment group students who enroll in any participating private school at any point within the four years and control group students who never enroll in any participating private schools. Non-compliance is represented by treatment group individuals who never use an LSP scholarship to attend a private school and control group students who enroll in any participating private schools at any point within the four years. Refer to Appendix B for additional details on treatment-control contrast.

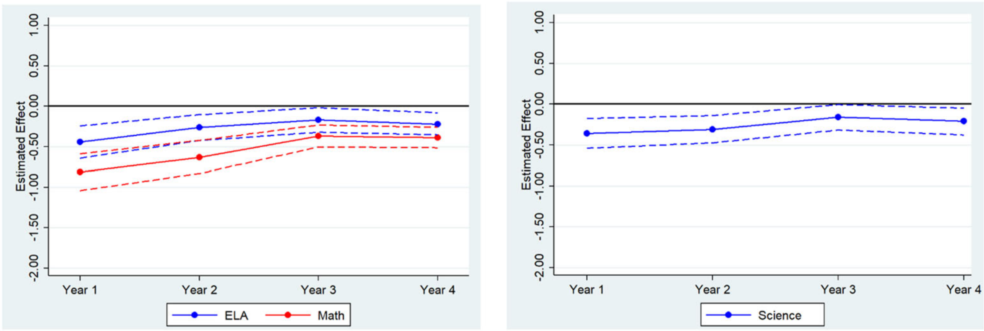

Figure 3. Estimated local average treatment effects in ELA, math, and science over time. Figure presents point estimates from fully specified models for Year 1 through Year 4 for ELA, math, and science. Results are presented for a consistent sample of students with Spring 2016 outcome data. ELA and math results are based on student achievement on the Louisiana state assessments in the school year 2012–13. Dashed lines represent 90% confidence intervals for the effect estimates. *Source*. Authors' calculations.

Our estimates of the impact of scholarship usage on student achievement begin with a simple model that only controls for first-choice school lottery fixed effects (column 2). Next, we include an indicator for taking the same grade-level assessment in two consecutive years due to grade-level retention (column 3). The fully specified models, which additionally control for student demographics (columns 4) are our preferred treatment effect estimates given the few significant differences in baseline characteristics between treatment and control groups observed in Table 5.

While previous research suggested that the initial, large negative test score impacts for LSP voucher users were declining over time, the estimated impact of LSP voucher usage presented in Table 5 is negative across all subject areas in fully specified models. The estimated effects are substantial, as LSP voucher users appear to score 22 percent of a standard deviation lower than control group students in ELA, 39 percent of a standard deviation lower in math, and 21 percent of a standard deviation lower in science (Table 5, Column 4). The negative achievement impacts are consistent and similar in magnitude when students in New Orleans are excluded from the sample (Table 5, Column 5).[24]

As an additional robustness test, we examine if differential attrition between treatment and control students affects the estimated causal effects. We find no significant evidence that differential attrition drives our findings.[25]

Next, we examine how the LSP effects vary over time for these samples of students. Figure 3 presents LATE estimates for ELA, math, and science for Years 1–4 for consistent samples of students contributing to the analyses presented in Table 5.[26] Impacts are

[24]These models attempt to provide a cleaner look at the effects of the program's statewide expansion because New Orleans was already home to both a diverse public charter school market (Harris & Larsen, 2018) as well as the pre-existing piloted version of the LSP.

[25]See Appendix B for the full differential attrition analysis.

[26]We present these figures of the impacts of the LSP on a consistent sample of students across time because the annual samples of students with outcome data change over the outcome time horizon. Focusing on a smaller but consistent sample of students means that we can rule out changing student populations as a factor affecting any changes in program effects observed across the years.

estimated by conditioning the ever enrollment variable to the possible period in which a student could have enrolled in a private school.[27]

We observe large declines in ELA, math, and science performance in the first year of voucher usage that become less negative in Years 2 and 3. Effects generally are the most negative in math. By Year 4, however, we observe a small direction reversal for the voucher usage effect estimates in all three subjects. Effects are slightly more negative in magnitude in Year 4 relative to Year 3 across all tests and samples and are statistically significant across all subjects. The longitudinal pattern of findings suggests that initial achievement losses due to participation in the LSP are substantially reduced but not fully eliminated after four years.

Student Achievement Subgroup Effects

The results presented in Table 6 allow us to determine if LSP voucher usage effects are differentiated by gender, race, and baseline achievement. Results are presented for both simple models which control only for LSP assignment lottery and fully specified models which additionally control for demographics. All models estimate the impact of using an LSP voucher to attend a private school at any point between 2012 and 13 and 2015 and 16.

In general, we do not observe consistent evidence that gender moderates LSP voucher usage effects. Difference estimates vary across models—sometimes positive and sometimes negative—while only one point estimate is statistically significant. In contrast, we observe evidence suggesting effects vary for students of different racial backgrounds. The causal effects of using a voucher are less negative for African American students relative to other LSP applicants, and estimated differences in impacts by race are statistically significant in all but one fully specified model (Table 6). Year 4 is the first time we have observed such LSP results.

Table 6 additionally includes analyses examining if there is an effect heterogeneity by baseline achievement category. The analytical sample for these analyses is restricted to students with both baseline and Year 4 outcome achievement, effectively reducing the sample to students applying for scholarships for grades four and five. While we largely do not observe differences by baseline achievement category, students scoring in the middle third of math achievement at baseline have very large and statistically significant negative effects by Year 4. We cannot, however, make direct comparisons across point estimates because these estimates are drawn from separate regressions. Interestingly, LSP students initially testing in the bottom third of the test distribution at baseline are shown, at times, to be outperforming their control group counterparts by Year 4, though none of the effects are statistically significant.

In short, while we do not observe strong evidence that effects were differentiated by gender or baseline achievement, our results indicate African American students experienced less negative achievement impacts from LSP voucher usage than non-African American students.

[27]For example, Year 2 effects are estimated for students who have complied with lottery assignment at some point during Year 1 and Year 2. Year 3 effects are estimated for students who complied with lottery assignment at some point during Year 1, Year 2, and Year 3, and so on.

Table 6. Differential effects of the LSP by gender, ethnicity, and baseline achievement.

	English language arts			Mathematics		
	N (1)	Simple (2)	Full model (3)	*N* (4)	Simple (5)	Full model (6)
Gender						
Female students	1182	−0.12	−0.22**	1173	−0.35***	−0.37***
		(0.10)	(0.10)		(0.09)	(0.10)
Male students	1028	−0.15*	−0.22*	1029	−0.29***	−0.40***
		(0.09)	(0.12)		(0.09)	(0.11)
Difference		0.03	0.00		−0.06	0.02
		(0.11)	(0.14)		(0.11)	(0.14)
Race/ethnicity						
Black students	1983	−0.09	−0.16*	1976	−0.29***	−0.34***
		(0.08)	(0.08)		(0.08)	(0.09)
Other students	227	−0.53**	−0.57**	226	−0.65***	−0.73***
		(0.23)	(0.24)		(0.22)	(0.20)
Difference		0.44*	0.41*		0.36	0.39*
		(0.23)	(0.24)		(0.25)	(0.23)
Baseline achievement						
Lower third	249	0.08	−0.10	245	0.25	0.20
		(0.33)	(0.28)		(0.22)	(0.22)
Middle third	254	0.05	−0.06	253	−0.78***	−0.82***
		(0.25)	(0.22)		(0.17)	(0.17)
Upper third	242	−0.16	−0.01	235	−0.40	−0.24
		(0.28)	(0.28)		(0.26)	(0.31)

Notes. ***$p < 0.01$, **$p < 0.05$, *$p < 0.10$. Performance measures standardized within-grade based on control group score distributions. Standard errors (parentheses) account for clustering within risk sets. First stage regressions indicate the LSP scholarship award result is a good instrument for actual use. *Baseline achievement* analysis requires students to have achievement scores prior to randomization (effectively restricting to students initially applying for grades fout and five). *Source*. Authors' calculations.

College Entrance Results

Next, we examine the impact of the LSP on students' likelihood of enrolling in college within six months following their expected high school graduation. Again, assuming we have identified random lotteries, the LATE estimates represent the causal effect of using an LSP scholarship on students' likelihood of entering college. The results of the LATE estimations are presented in Table 7. The first-stage estimated likelihoods of enrolling in an LSP participating private school by lottery status appear in Column 1. We find a relatively high take-up rate, with 77 percent of students who win a lottery for their first-choice private school using the awarded scholarship to enroll in a participating private school at any point between 2013 and 14 and 2015 and 16.

The second-stage results represent the causal effect of winning the first-choice lottery and ever enrolling in a participating LSP private school on subsequent college enrollment. Column 2 presents the results of the two-stage model without any student-level covariates. Column 3 displays results from our preferred models that include student-level covariates. We find no statistically significant causal effect of the LSP on college entrance for scholarship users. The LSP voucher users are 3.2 percentage points more likely to enter college than their control group counterparts, with an estimated 42.0 percent of treatment students enrolling in college within six months following their expected high school graduation compared to 38.8 percent of control students, a difference which is not statistically significant.

Next, we consider the impact of LSP usage on the likelihood of students enrolling in two- or four-year post-secondary institutions. The missions of the two types of higher

Table 7. Estimated effects of ever using an LSP voucher on college entrance.

	First stage	Late	
		Simple model	Fully specified
	(1)	(2)	(3)
College entrance	0.770***	0.031	0.032
	(0.025)	(0.034)	(0.034)
2-yr. Institution	0.839***	−0.027	0.033
	(0.031)	(0.036)	(0.036)
4-yr. Institution	0.839***	0.003	−0.002
	(0.031)	(0.030)	(0.030)
Controls			
Demographics	X		X
No. of school choices	X		X
N	1,113	1,113, 1,112	1,113, 1,112
Lotteries	106	106, 82	106, 82

Notes. ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Simple Models contain estimations that only control for first-choice school lottery fixed effects. Fully Specified models control for students' gender and race as well as the number of school preferences listed on applications. Bootstrap standard errors in parentheses account for clustering of students within lotteries. All models are linear probability models. Linear predictions of all models fall within the appropriate range of zero to one. Estimates on the probability of attending a two- or four-year institution are conditional on having entered any college. In columns 2 and 3, the first number presented for "*N*" and "Lotteries" represents the sample size and number of lotteries for the "College entrance" analysis. The second number represents the sample size and lottery number for the "2-yr. Institution" and "4-yr. Institution" models. *Source*: Authors' calculations.

education institutions tend to differ, as two-year colleges typically seek to prepare students for a trade or to transfer to a four-year college, while four-year colleges typically seek to coax students toward earning a bachelor's degree. Prior research, while limited, finds varying student enrollment patterns between two- and four-year institutions (Chingos et al., 2019; Wolf et al., 2019).

We estimate the effect of LSP usage on enrolling in a two- or four-year institution in separate models. Our results indicate that scholarship users enter two- and four-year institutions at similar rates as their control student counterparts. Scholarship users enroll at a slightly higher rate, by 3.3 percentage points, in two-year institutions and at a slightly lower rate, by 0.2 percentage points, in four-year institutions than their control counterparts, but these differences are statistically null (Table 7).

College Entrance Subgroup Results

The results in Table 8 show LSP voucher usage effects on college entrance differentiated by gender and race. Results are presented for fully specified models which control for assignment lotteries and student demographics. All models estimate the impact of using an LSP voucher to attend a private school at any point between 2012–13 and 2015–16 on the likelihood of students entering college within six months following their expected high school graduation. We find no evidence that voucher usage effects vary between female and male students or between African American students and students of other races.

Discussion

This study examines if the statewide expansion of the Louisiana Scholarship Program (LSP) affected student achievement and the likelihood of students enrolling in college.

Table 8. Subgroup effects of the LSP on college entrance.

	N (1)	Full model (2)
Gender		
Female students	579	0.037
		(0.049)
Male students	534	0.026
		(0.050)
Difference		0.011
		(0.073)
Race/Ethnicity		
African–American students	1008	0.050
		(0.035)
Other students	105	−0.120
		(0.114)
Difference		0.169
		(0.117)

Notes: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Full Model controls for students' gender and race as well as the number of school preferences listed on applications. Bootstrap standard errors in parentheses account for clustering of students within lotteries. All models are linear probability models. Linear predictions of all models fall within the appropriate range of zero to one. *Source*: Authors' calculations.

Using a lottery-based design, we estimate unbiased, causal impacts of LSP voucher usage on two important categories of student outcomes. We observe large, statistically significant, negative effects of the LSP on student achievement on the state accountability test after four years. In contrast, we find no significant causal effect on the rate at which LSP participating students enroll in post-secondary schooling. The college entrance results, while null, greatly enhance the developing literature on the attainment effects from private school choice programs because only five previous studies exist that measure post-secondary outcomes from two voucher programs in Milwaukee, Wisconsin, and in Washington, DC.

One might find it odd to see a voucher program associated with such large negative test score effects to not also decrease the likelihood of students entering college. Nevertheless, there are several probable explanations for the potential disconnect between the LSP's effects on student test scores and post-secondary enrollment. First, students in the achievement analysis are not the same group of students in the attainment analysis. The Year 4 achievement analysis includes students who applied for grades one through five in the first year of the program, 2012–13, while the college entrance sample includes students who applied that year for grades seven through twelve (Figure 2). The differences between the test score findings and the college enrollment findings could be due to the difference between the LSP experience of students in earlier or later grades.

Fortunately, we observe both Year 1 test score outcomes and college enrollment status for students who applied for grades seven or eight in the first year of the program. This allows for exploratory analysis to estimate the achievement and college enrollment effects for the same group of students. Using our fully specified models for each outcome, the effects of the LSP on achievement and attainment outcomes are less disconnected for seventh- and eighth-grade applicants than when comparing the effects from the full achievement sample to the full attainment sample. Seventh- and eighth-grade scholarship users experience no statistically significant achievement

Table 9. Estimated effect of LSP usage on student achievement in year 1 and college entrance for 7th- & 8th-grade applicants.

	Achievement sample			College enrollment sample	
	N (1)	Full model (2)		*N* (3)	Full model (4)
Panel A: english language arts			Panel D: college entrance		
Grade 7 or 8 applicants	530	−0.05	Grade 7 or 8 applicants	657	−0.013
		(0.15)			(0.050)
Grade 3–6 applicants	1771	−0.33***	Grade 9–12 applicants	456	0.080*
		(0.09)			(0.047)
Difference		0.28*	Difference		−0.093
		(0.17)			(0.067)
Panel B: mathematics					
Grade 7 or 8 applicants	529	−0.20			
		(0.17)			
Grade 3–6 applicants	1773	−0.64***			
		(0.09)			
Difference		0.44**			
		(0.2)			
Panel C: science					
Grade 7 or 8 applicants	523	−0.18			
		(0.17)			
Grade 3–6 applicants	1768	−0.33***			
		(0.08)			
Difference		0.15			
		(0.19)			

Notes: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Achievement outcomes are Year 1 standardized test score data. Performance measures standardized within-grade based on control group score distributions. Full Model controls for student demographics and the number of school preferences listed on applications. Standard errors (parentheses) account for clustering within risk sets. First stage regressions indicate the LSP scholarship award result is a good instrument for actual use.
Source: Authors' calculations.

or college entrance effects (Table 9). Considering the achievement sample for all test subject areas, we observe negative point estimates for LSP users applying for the seventh and eighth grades ranging from a 5 percent of a standard deviation decline in ELA to a 20 percent of a standard deviation decline in math, but none of the estimates are statistically significant (Panel A and B, Column 2). The treatment effect difference between seventh- and eighth-grade applicants and applicants to earlier grades is statistically significant in both ELA (Panel A) and math (Panel B), with the largest difference—44 percent of a standard deviation—in math achievement. The college entrance effect for seventh- and eighth-grade applicants is negative but small and statistically insignificant (Panel D). Conversely, scholarship users who entered high school in the first year of the LSP enter college within six months of their expected high school graduation at a rate that is 8 percentage points higher than their control counterparts (Panel D). However, the difference between the estimated effects for scholarship users who apply for seventh- or eighth-grade and those for ninth- through twelfth-grade falls short of statistical significance.

Our analysis essentially produced four distinct findings across three different grade ranges of students. For students who won a lottery for placement into private schools in the elementary grades of one through five in 2012–13, we find that their participation in the LSP led to lower scores on the state accountability tests in math, ELA, and science compared to their peers who lost a lottery and did not attend a participating private school. These effects are substantial, ranging from a decline in science scores by 21 percent of a standard deviation to a drop of 39 percent of a standard deviation in math scores. These elementary-age students were too young for us to assess the LSP's impact on their college enrollment rates.

For students who won an LSP lottery in the junior high school grades of seven and eight in 2012, we can estimate the effect of the voucher program on both achievement after one year and attainment. The one-year achievement effects of the LSP on a combined sample of baseline seventh- and eighth-grade students, while negative, are not significantly different from zero. The effect of LSP participation on students' rate of enrollment in a two- or four-year college within six months of their forecasted graduation from high school similarly is null.

Finally, for the older students placed in the LSP by lottery in the high school grades of nine through twelve, participation in the voucher program increased their rate of college enrollment by 8 percentage points, with an estimated 47.5 percent of treatment students entering college compared to an estimated 39.5 percent of their control counterparts. That impact amounts to a 20.2 percent increase in the likelihood of ninth- through twelfth-grade scholarship users entering college.

What might explain this pattern of results, which are unprecedented among studies of vouchers in the U.S.? The simplest explanation is that the private elementary schools participating in the LSP were of lower quality than the large public schools attended by the control group students, while the private high schools participating in the voucher program were of higher quality than the counterfactual public high schools, with the junior high schools attended by the LSP and control group students approximately equal in quality. We have some evidence to suggest this is the case. Of the LSP participating private schools that specialized in either elementary or high school grades, 27 percent of the elementary schools were sanctioned for unacceptable levels of performance under the Louisiana Department of Education accountability system while 17 percent of high schools were similarly sanctioned.[28] It is possible that the treatment-control contrast in school quality was negative for the younger LSP students but positive for the older ones, thus explaining, in part, the pattern of results.[29]

A second factor that varied across the grade ranges of the private schools in the LSP was the proportion of their 2012 enrollments consisting of new LSP students. For private schools in the program serving only students in the elementary grades, that proportion of new students joining their school community through the voucher program in 2012–13 averaged 29.5 percent. For the private schools in the LSP serving the high school grades, the average was just 14.9 percent. A large influx of new students from disadvantaged backgrounds can overwhelm a school logistically and challenge the existing school culture (Akerlof & Kranton, 2002). A separate study of the possible mediators

[28]We are only able to match school characteristics to a sample of 107 participating private schools. Private school characteristics came from the LDOE, the Private School Universe Study, as well as individual phone calls to the schools from members of our research team. Data on which schools were sanctioned were provided by LDOE and include all LSP participating schools and not just oversubscribed schools.

[29]We ran exploratory analysis looking at the achievement impacts for students in the first year who were in sanctioned vs non-sanctioned schools. We find negative achievement impacts for both groups, but the effect is less negative for students in non-sanctioned schools. This analysis is limited to only year 1 and relies on a smaller sample due to data limitations. Results available upon request.

of the achievement effects of the LSP after one and two years found that higher proportions of LSP students in a school's population were negatively associated with the achievement effects of the program (Lee et al., 2020). The relationship was only statistically significant in ELA (not math) and only for two-year achievement effects. Still, there is some suggestive evidence that the LSP programmatic effects may have been better for high school students than for elementary students because the high schools were less overwhelmed by the influx of new LSP students than were the elementary schools.

While variation in the treatment-control contrast regarding school quality and differences in the proportion of the student body composed of new LSP students are plausible standalone explanations for the complete pattern of four LSP findings we present here, other factors in combination could explain differences among the individual findings. First, the negative achievement results could reflect a lack of alignment of private school curricula with the Louisiana state standards rather than the negative impacts of LSP schools on actual student learning. There is some suggestive evidence for this hypothesis. We analyzed achievement using student performance on Louisiana state criterion-referenced exams aligned with Louisiana's state public school academic standards. In the few years and grades where a less-aligned state test was administered, the negative achievement impacts of the LSP were smaller and, in some cases, not statistically significant (Mills & Wolf, 2017a). An in-depth investigation through 2015, the outcome year of our achievement analysis, found only one Louisiana private school that had adopted the entire set of Common Core Standards on which the state criterion-referenced tests are based (McElfresh, 2015). It is possible that a more general measure of learning than the state criterion-referenced accountability exam would have revealed LSP achievement effects that were more favorable, or at least less unfavorable, to the program than those reported in the test score analysis for elementary students.

Second, the negative achievement results could reflect a public school advantage working with high-stakes accountability policies. Our achievement analysis is limited by two assessment regime changes, and thus three different tests, implemented over the four-year period. While we can address the problem of changing tests in part through standardizing test scores within each year, test changes can have disruptive impacts on both students and schools. In Year 3 of our outcome analysis, when Louisiana adopted the PARCC as its accountability exam for the first time, both the public schools and the private schools in the LSP were given a hiatus from any consequences from the performance of their students. For that one year, the accountability test was low-stakes for all schools, and the scores of the LSP and control group students were estimated to be statistically similar (Mills & Wolf, 2017b). When the accountability test was switched back to a modified version of the LEAP/iLEAP exam the next year, and the stakes for the schools were once again high, we observed the achievement effects of the LSP in that fourth evaluation year to be significantly negative. It is possible that Louisiana public schools, with more than a decade of experience preparing students to do well on accountability tests, hold an institutional advantage over Louisiana private schools in generating better student test scores when the stakes are high. Moreover, private schools, for which the administration of high-stakes accountability tests is a new activity, likely experience more disruption from a change in the testing regime than public schools with more high-stakes testing experience. While we cannot know for sure the extent to which different levels of experience with high-stakes testing explains the negative test score effects we observe for LSP elementary students, we can say, conclusively, that four years of experience administering three different state accountability tests did not enable the LSP elementary schools to generate test scores for their LSP students that were as high as those of students in the control group, with the exception of the third year, when the test was both new and low-stakes.

Third, the set of null effects of the LSP on both the achievement and attainment of students entering the seventh or eighth grades in the fall of 2012 could be due to low study power for that subgroup. Only 657 students in our analytic sample composed the subsample of seventh- and eighth-grade students for which we observed at least some initial achievement effects in addition to effects on college enrollment rates. Still, an even smaller subsample of study participants, 456, were entering grades nine through twelve at baseline. That even lower-powered sample demonstrated a positive LSP effect on college enrollment that was at least marginally statistically significant. Moreover, the subsample of baseline seventh- and eighth-grade students did not generate statistically significant negative effects in the first outcome year of our study, when the complete sample of tested students produced large and statistically significant negative LSP effects. Our exploratory subgroup analysis confirmed that the test-score effects of the LSP on the seventh- and eighth-grade baseline sample were statistically different from the program's achievement effects on the younger subgroup of participants. For these reasons, we think that our achievement and attainment findings for the students in grades seven and eight are true nulls and not type II errors.

We also think that our finding of a positive LSP attainment effect for students who started the program in the high school grades, though only marginally statistically significant, is a reliable finding. Although it could be an artifact of the private schools in the LSP applying less stringent standards for high school graduation than the standards of the public schools enrolling control group students, we see no evidence indicating that LSP students were more likely to enroll in less selective institutions. Given these similarities in the quality of the colleges attended by the two groups, the 8 percentage point higher enrollment rate for the LSP students who entered the program in high school appears to be a truly positive effect of the voucher program on education attainment. It is possible that private schooling could have a different effect on students' later attainment if experienced in high school compared to earlier grades (Wolf et al., 2019).

Conclusion

Leveraging the assignment of scholarships by lotteries, our evaluation of the Louisiana Scholarship Program finds substantial negative impacts on student achievement in the first year of the program that diminish yet persist into the fourth year of the program. In addition, we find no significant effect of scholarship usage on the likelihood of students entering college in our full analytic sample. For ease, Table 10 includes a summary of all estimated treatment effects for scholarship users and their control counterparts. Ours is the first evaluation of a statewide school voucher program to include both achievement and attainment outcomes. It is also the

Table 10. Summary of LSP effects for treatment and control students across outcomes.

	Regression adjusted LSP scholarship users avg. (1)	Control group compliers avg. (2)	Estimated effect (3)	*p*-value (4)	*N* (5)
Achievement outcomes after four years					
English language arts	−0.15	0.06	−0.22***	0.009	1632
Mathematics	−0.30	0.09	−0.39***	0.000	1625
Science	−0.11	0.10	−0.21**	0.034	1606
College entrance effects					
Enroll	0.349	0.317	0.032	0.345	1113
2-yr. institution	0.211	0.178	0.033	0.354	1112
4-yr. Institution	0.137	0.139	−0.002	0.944	1112

Notes: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Achievement outcomes are Year 4 standardized test score data. Performance measures standardized within-grade based on control group score distributions. "Control group complier"" constitutes the year 4 average outcome for students who lost their first-choice lottery and never enrolled in a private school. "Estimated effects" come from the full regression models with controls for student demographics and the number of school preferences listed on applications (column 4 of Table 5 and column 3 of Table 7). "LSP scholarship users" constitute the year 4 average outcome for students who won their first-choice lottery and enrolled in a participating private school. Following Wolf et al. (2013), these averages are calculated by adding the "Estimated effect" in column (3) to the "Control group avg." in column (2). Numbers may not match exactly due to rounding. *Source*. Authors' calculations.

first evaluation of any U.S. voucher program to examine its effects on science achievement.

Our estimates of the impact of using an LSP voucher are causal because they are grounded in the random assignment of students for placement in their first-choice private school. While this evaluation has high internal validity, the Louisiana public and private education landscape, as well as the design of the LSP, are important to consider when generalizing these results to broader school choice policies.

While acknowledging the limitations of our study, we think that it presents lessons for both education policy and future research on school vouchers. Regarding the adoption, design, and implementation of private school voucher programs, it is a cautionary tale. For those who hold that scores on state accountability tests are the sole or primary measure by which education policies should be evaluated, our findings join those from recent evaluations of the Indiana and Ohio voucher programs in arguing against the desirability of statewide school choice initiatives. Policy makers might instead consider concentrating voucher programs to urban areas with a rich set of private schools from which parents can choose. Previous evaluations of such voucher programs in Cleveland, Milwaukee, and the District of Columbia report achievement effects that are more positive than those we find for the statewide LSP.

A second policy lesson from our findings is that legislators should consider carefully the implications of requiring that participating private schools administer state criterion-referenced accountability tests. Such assessments are designed to measure the extent to which a student has mastered the public school curriculum matched to their specific grade level. To the extent that private schools provide a curriculum that differs from the public school one, in the specific content or sequencing of concepts, differences in scores on the state test between voucher and comparison students will reflect some unknown combination of curricula differences and achievement differences. Moreover, if private schools align their curricula to the state standards, those schools will present fewer distinctive choices to parents. Perhaps reflecting the desire of private school leaders to avoid such a Faustian bargain, two survey experiments demonstrate that leaders report being less willing to participate in a hypothetical voucher program if their private schools will be required to administer the state test (DeAngelis et al., 2019, 2020). The LSP had a low private school participation rate of 33 percent, which may have contributed to the negative achievement effects it produced (Abdulkadiroğlu et al., 2018; Sude et al., 2018). Generally, policy makers should consider how school voucher regulations placed on private schools could limit the quality of school options available to participating families.

Finally, regarding implementation, in our opinion, no state should again attempt to fully launch a new private school voucher program in less than 90 days, as Louisiana did with the LSP. Some private school personnel showed up in August to prepare for the start of the 2012–13 school year not even knowing that dozens, in some cases even hundreds, of new students would be joining their educational communities. The largest negative effects of the LSP on student achievement were observed in that first outcome year. We suspect that the rushed implementation of the program was a contributing factor.

Our study also offers lessons for future research. First, given the close correlation between the type of test used and the resulting achievement effects reported in voucher evaluations, a single study that identifies the test-score effects of a voucher initiative using both state criterion-referenced exams and a norm-referenced assessment on the same sample of students would be especially welcome. Such a within-study comparison using alternative assessments would signal how much, if any, of the negative achievement findings from recent voucher evaluations are attributable to the specific test being used and how much, if any, are due to actual learning losses caused by participation in the voucher program.

Second, evaluations of sector-wide education interventions such as private school voucher programs should employ measures of multiple relevant student outcomes. We expect elementary and secondary schools to impart learning to students, keep them safe, place them on a path to graduate from high school and obtain higher education or gainful employment, shape the social and emotional skills of their charges, and prepare them for the civic responsibilities they will face as adults. Schools likely face tradeoffs in their focus on one, several, or all these valued educational outcomes. Voucher evaluations will provide a more complete picture of the effectiveness and desirability of private school choice programs to the extent that they determine voucher effects on as many of these outcomes as possible. We have sought to do so here, including voucher effects on achievement in three specific domains as well as initial attainment, but more could and should be done.

More school voucher evaluations should include the scope and scale to power analyses of the moderators and mediators of voucher effects. Finally, more research is needed on the interplay between the design and context of private school voucher programs, on the one hand, and the number, quality, and diversity of private schools that choose to participate, on the other hand. If anything, our study suggests that a private school voucher program's success can depend heavily on its design.

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Appendix A: Comparing Characteristics of Applicants in Over-Subscription Lotteries with Other Applicants

Table A1 examines the extent to which LSP applicants contributing to our analysis differ from other qualified LSP applicants who did not experience over-subscription placement lotteries. Specifically, we present linear probability models predicting the likelihood of a qualified LSP applicant participating in an oversubscription lottery for their first-choice school. All analyses focus on eligible applicants who did not receive exclusions in the LSP placement algorithm (students with disabilities, multiple birth siblings, and applicants who formerly attended "A" graded schools). Models 1 and 2 include student-level characteristics. Model 3 focuses on the LSP algorithm's priority categories (some of which factor in student characteristics such as New Orleans pilot program participation, etc.).[30] Model 1 does not control for a geographic region, while all remaining models include residency ZIP code fixed effects.

The results presented in Table A1 indicate several significant differences between over-subscription lottery participants and other LSP applicants. Specifically, NOLA pilot program participants are less likely to be in lotteries while students who offered more preferences were more likely to be in lotteries than those who offered one school and students in priority categories 4 and 5 (new to the system effectively) were more likely to be in lotteries. These relationships are largely consistent across models and do not depend strongly on the inclusion or exclusion of ZIP code fixed effects. Taken together, these findings are a reminder of the limits one should place on the generalizability of our findings. Specifically, our focus on producing internally valid estimates of the causal effect of LSP scholarship usage on student academic measures restricts our ability to generalize these findings to all LSP participants.

[30]LSP algorithm priority categories: (1) students who received LSP scholarships in the prior school year who are applying to the same school; (2) non-multiple birth siblings of Priority 1 awardees in the current round; (3) students who received LSP scholarships in the prior school year who are applying to a different school; (4) new applicants who attended public schools that received a "D" or "F" grade in Louisiana's school accountability system at baseline; (5) new applicants who attended public schools that received a "C" grade.

Table A1. Student characteristics predicting the likelihood of being in an oversubscription lottery for their first-choice school.

	(1)	(2)	(3)
Female	0.0042	0.0064	0.0026
	(0.0112)	(0.0112)	(0.0116)
Student race/ethnicity (comparison: black students)			
Hispanic	−0.0027	−0.0155	−0.0090
	(0.0454)	(0.0419)	(0.0401)
White	−0.0974***	−0.0394	−0.0401
	(0.0344)	(0.0301)	(0.0301)
Other	−0.0305	−0.0316	−0.0238
	(0.0279)	(0.0282)	(0.0275)
Free- or reduced-price lunch	−0.0260	−0.0208	−0.0232
	(0.0251)	(0.0218)	(0.0239)
New Orleans pilot program	−0.3878***	−0.2699***	
	(0.0776)	(0.0758)	
School letter grade (comparison: B school)			
C	−0.2795***	−0.2996***	
	(0.0796)	(0.0775)	
D	0.0604	0.0593	
	(0.0781)	(0.0784)	
F	0.1212	0.1177	
	(0.0818)	(0.0808)	
Total school preferences listed (comparison: 1 choice)			
2	0.0935***	0.0684***	0.0736***
	(0.0192)	(0.0167)	(0.0177)
3	0.1354***	0.0880***	0.0895***
	(0.0241)	(0.0248)	(0.0247)
4	0.1262***	0.0758***	0.0762***
	(0.0250)	(0.0232)	(0.0232)
5	0.0944***	0.0537**	0.0526**
	(0.0254)	(0.0233)	(0.0222)
LSP algorithm priority category (comparison: category 1)			
2			−0.3690***
			(0.0200)
3			−0.1439**
			(0.0599)
4			0.3130***
			(0.0358)
5			−0.0691
			(0.0425)
Student grade of application (comparison 4th)			
1st	0.1010**	0.0971**	0.0998**
	(0.0392)	(0.0412)	(0.0411)
2nd	0.0099	0.0039	0.0029
	(0.0317)	(0.0315)	(0.0327)
3rd	0.0405	0.0360	0.0279
	(0.0360)	(0.0373)	(0.0376)
5th	0.0441	0.0338	0.0364
	(0.0390)	(0.0402)	(0.0388)
6th	0.0473	0.0363	0.0350
	(0.0389)	(0.0391)	(0.0378)
7th	−0.0065	−0.0246	−0.0203
	(0.0259)	(0.0253)	(0.0258)
8th	−0.1142***	−0.1577***	−0.1561***
	(0.0330)	(0.0338)	(0.0344)
9th	0.0009	−0.0139	−0.0135
	(0.0367)	(0.0352)	(0.0356)
10th	0.0408	0.0045	0.0020
	(0.0608)	(0.0591)	(0.0595)
11th	0.0427	0.0285	0.0320
	(0.0836)	(0.0785)	(0.0796)
12th	−0.1217	−0.1630*	−0.1666*

(*continued*)

Table A1. Continued.

	(1)	(2)	(3)
	(0.0964)	(0.0858)	(0.0873)
Constant	0.6874***	0.6778***	0.4582***
	(0.0834)	(0.0782)	(0.0383)
Observations	5,954	5,954	5,954
R-squared	0.2489	0.3459	0.3669
Zip Code FE	No	Yes	Yes

Notes: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. Linear Probability Models predicting the likelihood of a qualified LSP applicant participating in an oversubscription lottery for their first-choice school. All analyses focus on eligible applicants who did not receive exclusions in the LSP placement algorithm (students with disabilities, multiple birth siblings, and applicants who formerly attended "A" graded schools). Models 1 and 2 include student-level characteristics. Models 3 focuses on the LSP algorithm's priority categories (some of which factor in student characteristics such as NOPP participation, etc.). Standard errors account for the clustering of students within zip codes. *Source*: Authors' calculations.

Appendix B: Student Enrollment Patterns

Treatment-Control Contrast in the Achievement Sample

While eligible applicants were randomly assigned to receive or not to receive an LSP scholarship to their most preferred private school, participating families were not required to use the scholarship. Therefore, it is important to verify that treatment assignment is strongly correlated with school sector enrollment. Table A1 describes the patterns of enrollment for student applicants for the 2012–13 LSP cohort who received and who did not receive LSP scholarships to their first-choice schools for the four years following their initial application to the program. The analytical sample presented in Table A1 reflects students who did not list a special education classification on their LSP application, and who were not multiple birth siblings. Because our LATE analysis focuses on the results of first-choice school lotteries, the control group includes students who were never awarded a scholarship and students who received a scholarship to one of their non-first-choice private school preferences. The latter group, accounting for between 7 and 14 percent of control group students across all four years, are control-group crossovers in our LATE analysis.

While the majority of lottery winners used their scholarships to attend private schools, around 75 percent of students who did not receive scholarships attended public-sector schools in all years of our study. The percentage of first-choice lottery winners attending private schools has declined over time from 77 percent in Year 1 to only 41 percent by Year 4. More importantly, Year 4 is the first year in which we observe a larger percentage of first-choice school lottery winners attending public schools (traditional, charter, or magnet) than private schools (44 percent and 41 percent, respectively).

Differential Attrition in the Achievement Samples

We examine if differential attrition between treatment and control group members is impacting our results. Table A1 indicates noticeable sample attrition that has generally increased over time. Moreover, attrition rates appear to differ between treatment and control group members, with generally higher observed attrition rates among those initially assigned to the LSP control group. These differences, if driven by nonrandom factors, raise concerns of biased effect estimates (Gerber & Green, 2012; Lee, 2009; What Works Clearinghouse, 2014). If, for example, those in the control group with lower expected outcomes both in public and private schools leave the sample with a higher probability, our LATE estimates will be negatively biased.

In Table A2, we examine the extent to which differential attrition impacts our estimates using a bounding procedure Lee (2009) developed. If one knows nonrandom attrition is concentrated solely in either the treatment or control group, Lee (2009) shows that the true, unbiased program effect lies between two bounds created by parsing away the top and bottom performers from the non-affected group. Lee's (2009) bounding method relies on two assumptions: the assignment mechanism is random and sample selection is a monotonic function of treatment status. The LSP lottery process satisfies the first assumption. The second assumption requires that there are no LSP applicants who were assigned an LSP scholarship but decided to forgo their scholarship and instead enroll in a private school at their own expense. While we cannot validate this assumption empirically, it seems unlikely such "defiers" exist in our data—especially given the program's income threshold.

In our case, we are concerned about disproportionately high levels of attrition among low performers in the control group. Using Lee's method, we produce an upper bound of the true effect

Table B1. School enrollment patterns by scholarship award for achievement sample.

	Treatment group (received LSP to first-choice school)		Control group (did not receive LSP to first-choice school)	
	N	%	*N*	%
Year 1 (2012–13)				
Private school	493	77%	58	7%
Public school	113	18%	678	82%
Unknown/missing school	31	5%	91	11%
Year 2 (2013–14)				
Private school	552	60%	147	14%
Public school	270	30%	777	74%
Unknown/missing school	93	10%	132	13%
Year 3 (2014–15) – PARCC data				
Private school	684	52%	157	11%
Public school	477	37%	1036	75%
Unknown/missing school	145	11%	189	14%
Year 4 (2015–16)				
Private school	541	41%	127	9%
Public school	599	46%	1065	77%
Unknown/missing school	166	13%	190	14%

Notes: All students participated in LSP lotteries. Analysis sample excludes students with disabilities and multiple birth siblings. Year 1 is restricted to students applying for grades three through eight for the 2012–13 school year. Year 2 is restricted to applicants for grades two through seven. Year 3 is restricted to applicants for grades one through six. Year 4 is restricted to applicants for grades one through five. *Source*: Authors' calculations.

Table B2. Examining effects of differential attrition for the achievement sample.

	N (1)	Simple (2)	Full model (3)
Panel A: English language arts			
Primary LATE	2210	−0.13*	−0.21**
		−0.08	−0.08
Lee lower bound	2190	−0.19**	−0.27***
		−0.08	−0.08
Lee upper bound	2190	−0.04	−0.13
		−0.07	−0.08
Panel B: Mathematics			
Primary LATE	2202	−0.32***	−0.38***
		−0.07	−0.08
Lee lower bound	2181	−0.37***	−0.43***
		−0.07	−0.08
Lee upper bound	2181	−0.25***	−0.32***
		−0.07	−0.07
Controls			
Test re-take			X
Baseline achieve.			X
Demographics			X
No. of school choices			X
New Orleans			X

Notes: ***$p < 0.01$; **$p < 0.05$; *$p < 0.10$. "Simple Model" refers to estimations that only control for first-choice school lottery fixed effects. "Full Model" refers to models controlling for test retaking, student demographics, the number of school preferences offered, and geography. Performance measures standardized within-grade based on control group score distributions. Standard errors (parentheses) account for clustering within lotteries. First stage regressions indicate the LSP scholarship award result is a good instrument for actual use. *Source*: Authors' calculations.

by re-estimating the LATE effects on a subsample that excludes the lowest Year 4 performers in the treatment group. Similarly, we create a lower bound by excluding the highest performers. This method provides valid bounds on the true LATE by assuming that all differential non-responders in the control group were either the highest performers (lower-bound estimate) or lowest performers (upper-bound estimate) among all applicants.

Table 2A presents little evidence which supports the claim that our effect estimates are biased by differential attrition. Nearly all point estimates indicate that LSP voucher usage negatively impacted student achievement, especially in math. Even when they are not statistically significant, we observe negative effect estimates.

Appendix C: Baseline Achievement Analysis Sample

While our previous analyses controlled for baseline achievement (Mills, 2015; Mills & Wolf, 2017a; 2017b), the primary analyses presented in this article do not as this would restrict our analysis to applicants for grades four and five only. Moreover, dropping the baseline achievement requirement in no way impacts the internal validity of our estimates, which is derived from the lottery-based random assignment of treatment rather than our ability to control for baseline achievement.

To test the robustness of our primary results and provide a link to our previous research, we repeat the analysis on the smaller sample with baseline test scores (Table B1). Similar to the primary results presented in Table 2, we observe moderate rates of compliance across models and consistent statistically significant negative impacts in math for this smaller sample of students with baseline achievement. Specifically, LSP voucher users scored 28 percent of a standard deviation lower than their control-group counterparts in math by Year 4 (Table 2, Column 9). We additionally observe a statistically significant negative impact of LSP voucher usage on science test scores of 31 percent of a standard deviation. Point estimates for ELA are not statistically significant; however, this null finding may be due to a lack of precision as the standard errors are relatively large. Estimates that exclude students in New Orleans (Table B1, Column 5) are similar

Table C1. Estimated effects of ever using an LSP scholarship on student achievement after four years, baseline achievement sample.

		LATE			
	First stage (1)	Simple model (2)	+Test retake (3)	Fully specified (4)	Omitting new Orleans (5)
English language arts	0.61***	−0.12	−0.08	−0.11	−0.09
	(0.04)	(0.18)	(0.17)	(0.15)	(0.15)
Mathematics	0.61***	−0.27**	−0.25**	−0.28**	−0.30***
	(0.04)	(0.12)	(0.12)	(0.11)	(0.11)
Science	0.60***	−0.27	−0.24	−0.31**	−0.23
	(0.04)	(0.19)	(0.19)	(0.15)	(0.14)
Controls					
Test re-take			X	X	X
Baseline achieve.				X	X
Demographics				X	X
No. of school choices				X	X
New Orleans				X	
N	719−735		719−735	699−715	538−552
Lotteries	103−103		103−103	103−103	73−74

Notes. ***$p < 0.01$, **$p < 0.05$, *$p < 0.10$. *Simple Model* refers to estimations that only control for first-choice school lottery fixed effects. *Test Retake* indicates models including an indicator for if a student took the same subject test in two consecutive years. *Full Model* refers to models controlling for test retaking, baseline achievement, student demographics, number of school preferences offered, and geography. Column (5) omits students who attended New Orleans public schools in 2011–12, to account for the existence of the New Orleans pilot program. Performance measures are standardized within-grade based on control group score distributions. All models include first-choice school lottery fixed effects. Standard errors (parentheses) account for clustering within lotteries. First stage F-statistics all exceed Staiger and Stock's (1997) recommended threshold of 10. *Source*. Authors' calculations.

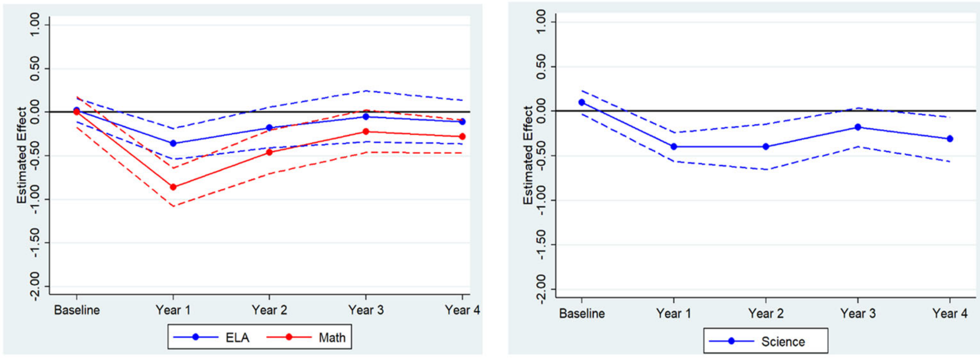

Figure C1. Estimated Local Average Treatment Effects in ELA, Math, and Science Over Time for Baseline Achievement Sample. *Notes*: Figure presents point estimates from fully specified models for baseline through year 4 for ELA, math, and science. Results are presented for a consistent sample of students with Spring 2016 outcome data. ELA and math results are based on student achievement on the Louisiana state assessments (LSA) in the school year 2011–12 through 2013–14, PARCC assessments in 2014–15, and LSA in 2015–16. Science results are based on student achievement on theLSA in 2011–12 through 2015–16. Dashed lines represent 90% confidence intervals for the effect estimates. *Source*: Authors' calculations.

to those from our preferred model (Table 2, Column 5) except the size of the negative impact in science drops to 23 percent of a standard deviation. These results likely suffer from low statistical power, a result driven in part by the requirement of baseline achievement, which effectively restricts our analysis to just two grade cohorts of LSP applicants.

Next, we examine how LSP effects vary over time for this subsample of students with baseline achievement. Figure B1 presents LATE estimates for ELA, math, and science for Years 1–4 for consistent samples of students contributing to the analyses presented in Table B1. As with Figure 3 in our primary analysis, impacts are estimated by conditioning the ever enrollment variable to the possible period in which a student could have enrolled in a private school.

Consistent with our primary analysis, we observe large declines in ELA, math, and science performance in the first year of voucher usage that become less negative in Years 2 and 3. Effects generally are the most negative in math. By Year 4, however, we observe a direction reversal for the voucher usage effect estimates in all three subjects, with statistically significant negative effects observed for both math and science. These results confirm the findings presented in the primary analysis that LSP scholarship usage does not fully reduce the large negative effects of voucher usage observed in the first year of program participation.